Divide decimals by whole numbers

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________ Date: ______________

1. Read the model: Calculate 28.56 divided by 8. Copy or complete the first step.

2. A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?

3. Calculate 34.75 + 8.906.

4. Miro says: Miro lines up the final digits instead of the decimal points. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate 28.56 divided by 8. Copy or complete the first step.
   Answer 3.57.

2. A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?
   Answer 8.975 km.

3. Calculate 34.75 + 8.906.
   Answer 43.656.

4. Miro says: Miro lines up the final digits instead of the decimal points. Is this correct? Explain.
   Answer Align ones with ones and decimal points vertically so each place-value column stays correct.

Divide decimals by whole numbers

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. Calculate 28.56 divided by 8.

2. A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?

3. Calculate 34.75 + 8.906.

4. Solve this independently and show every stage: Calculate 28.56 divided by 8.

5. Use a second method or representation for: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?

6. Check this result using an inverse, estimate or known fact: Calculate 34.75 + 8.906.

7. Prove or explain your reasoning: Calculate 50 - 17.485.

8. Identify and correct this misconception: Miro lines up the final digits instead of the decimal points.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Calculate 34.75 + 8.906.

Teacher answers and checking prompts

1. Calculate 28.56 divided by 8.
   Answer 3.57.

2. A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?
   Answer 8.975 km.

3. Calculate 34.75 + 8.906.
   Answer 43.656.

4. Solve this independently and show every stage: Calculate 28.56 divided by 8.
   Answer 3.57.

5. Use a second method or representation for: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?
   Answer Expected result: 8.975 km. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Calculate 34.75 + 8.906.
   Answer Expected result: 43.656. A valid check must be shown.

7. Prove or explain your reasoning: Calculate 50 - 17.485.
   Answer 32.515. The explanation must show why the method works.

8. Identify and correct this misconception: Miro lines up the final digits instead of the decimal points.
   Answer Align ones with ones and decimal points vertically so each place-value column stays correct.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Calculate 34.75 + 8.906.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Divide decimals by whole numbers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ________________

1. Prove or explain your reasoning: Calculate 50 - 17.485.

2. Prove the answer to this question, rather than only calculating it: Calculate 28.56 divided by 8.

3. Solve this in two different ways and compare them: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?

4. Work backwards from the answer to reconstruct the question: 43.656.

5. Explain why this reasoning fails: Miro lines up the final digits instead of the decimal points.

6. Create a new example that tests the same idea as: Calculate 34.75 + 8.906.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate 50 - 17.485.
   Answer 32.515. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Calculate 28.56 divided by 8.
   Answer Expected result: 3.57. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?
   Answer Expected result: 8.975 km. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 43.656.
   Answer One valid reconstruction is: Calculate 34.75 + 8.906. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro lines up the final digits instead of the decimal points.
   Answer Align ones with ones and decimal points vertically so each place-value column stays correct.

6. Create a new example that tests the same idea as: Calculate 34.75 + 8.906.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.